

9.2.12. Superset Device Types

A superset device type is a device type that is functionally similar or identical to a subset device type and has an overlap in its cluster requirements with a subset device type, subject to the following constraints:

- All clusters in the subset device type with mandatory conformance SHALL be supported in the superset device type with mandatory conformance.
- Any clusters in the subset device type with optional conformance MAY be supported in the superset device type.
- Any clusters in the subset device type with optional conformance that are also supported in the superset device type MAY have either optional or mandatory conformance in the superset device type.
- Element requirements of overlapping clusters in the superset device type SHALL be the same as in the subset device type.
- An endpoint implementing a superset device type MAY include the subset device type(s) in the DeviceTypeList of the descriptor cluster of the endpoint.
- There is a use case (rationale) for exposing the relationship between the superset and subset device type.

When these constraints apply, the subset device type SHALL be indicated in the "Superset Of" column of the table in the "Classification" section of superset device type, and guidance for server and client makers MAY be added.

9.2.12.1. Superset Device Types with identical cluster requirements

A special case of Superset/Subset device types is where the cluster requirements (as expressed in the constraints above) of two device types are identical. This typically occurs in cases where a new device type is introduced for a specific use case which is different from the use case of the existing device type, e.g. mounted on/off control is added for the in-wall use case versus the existing on/off plug-in unit. If there is a rationale to define one of these device types as the superset of the other, this SHALL be indicated in the "Classification" section of superset device type, and guidance for server and client makers SHOULD be added - also in the definition of the subset device type to hint readers to the superset device type with identical cluster requirements.

This guidance to implementers of servers and clients on how to use one or both device types SHOULD help to realize optimum representation in both 'new' clients, which understand both the existing and the new device type, and 'old' clients, which understand only the existing device type.

9.2.12.2. Superset Device Types - usage in Server

The endpoint exposing the server side of the clusters of the superset device type SHALL include the superset device type and MAY include the subset device type(s) in its DeviceTypeList attribute.

Example for Dimmable Light and Extended Color Light device types (Extended Color Light is a superset of Dimmable Light): An endpoint implementing device type Extended Color Light SHALL include device type Extended Color Light and MAY include device type Dimmable